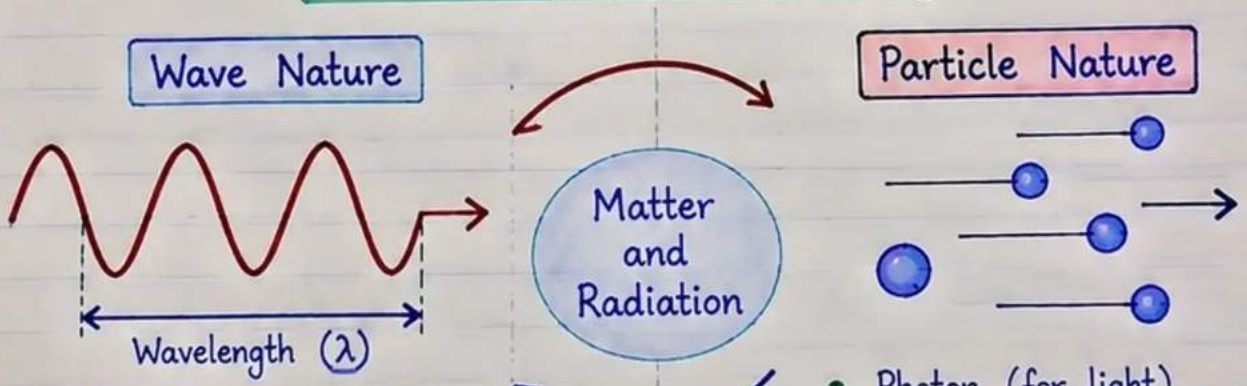
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Quantum Mechanics

Complete Handwritten Notes

For
BCA, MCA
B.Tech
All CS Students

Wave - Particle Duality



- Interference
- Diffraction
- Superposition

- Photon (for light)
- Electron (matter wave)
- Discrete packets of energy

- ☑ Complete Theory with Derivations
- ☑ Diagrams & Examples
- ☑ Numerical Problems
- ☑ Important Questions for University Exams

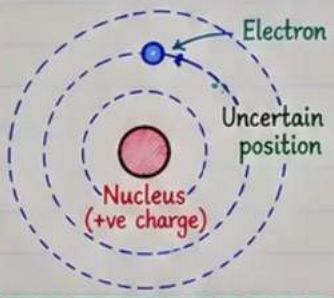
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5. Applications of Uncertainty Principle

Heisenberg Uncertainty Principle has many important applications in physics and modern technology. It shows that we cannot know both the exact position and momentum of a particle simultaneously. This principle helps in understanding the behaviour of microscopic particles and various physical phenomena.

1. Atomic Stability

- According to classical physics, an electron revolving around the nucleus should lose energy and spiral into the nucleus due to radiation, making the atom unstable.
- But due to uncertainty principle, the exact position and momentum of electron cannot be known, so electron cannot have zero radius.
- This prevents the electron from falling into the nucleus and explains the stability of atom.



Uncertainty principle prevents the electron from collapsing into the nucleus, ensuring the stability of atom.

2. Finite Size of Nucleus

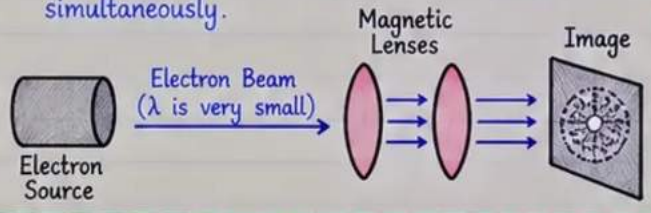
- If a particle (nucleon) is confined within a very small region (nucleus), then its momentum becomes uncertain and hence large.
- This explains why nucleons inside a nucleus have high kinetic energy and the nucleus has a finite size.

$$\Delta p \approx \frac{h}{\Delta x}$$

$\Delta x \approx 10^{-15}$ m (size of nucleus)
$\Rightarrow \Delta p$ is very large
$\Rightarrow$ High kinetic energy

3. Electron Microscope

- In an electron microscope, electrons are used instead of light to get higher resolution.
- Due to small de Broglie wavelength of electrons, we can observe very small objects.
- However, uncertainty principle limits the resolution because the exact position and momentum of electron cannot be known simultaneously.

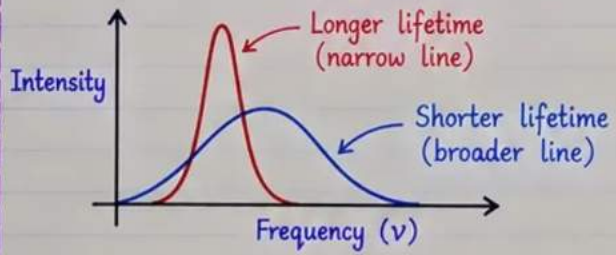


4. Spectral Line Broadening

- The uncertainty in energy and time is given by :

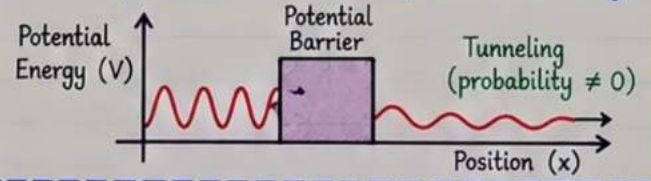
$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

- For a short-lived excited state (small Δt), the uncertainty in energy (ΔE) is large.
- This leads to broadening of spectral lines.



5. Quantum Tunneling

- Uncertainty principle allows a particle to have a range of momentum and energy.
- Even if the particle's energy is less than the potential barrier, there is a finite probability of finding the particle on the other side.
- This phenomenon is called quantum tunneling.



6. Zero-Point Energy

- Due to uncertainty principle, a particle cannot have both zero position and zero momentum.
- Hence, its kinetic energy can never be zero, even at absolute zero temperature.
- This minimum energy is called zero-point energy.

$$E_0 = \frac{1}{2}\hbar\omega \quad \text{(for harmonic oscillator)}$$

7. Other Applications

1. Explains the stability of matter and atoms.
2. Helps in understanding chemical bonding.
3. Important in semiconductor physics and nanotechnology.
4. Used in modern devices like STM (Scanning Tunneling Microscope).
5. Explains why we cannot have a perfectly precise measurement at microscopic level.

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2. de Broglie Matter Waves & Wavelength

de Broglie Hypothesis :

- In 1924, **Louis de Broglie** proposed that all moving material particles (matter) have wave nature, just like light.
- These waves associated with material particles are called **matter waves** or **de Broglie waves**.
- It explains the wave-particle duality of matter.

Louis de Broglie



"Just as light has wave natture, material particles also have wave nature."

de Broglie Wavelength (λ) :

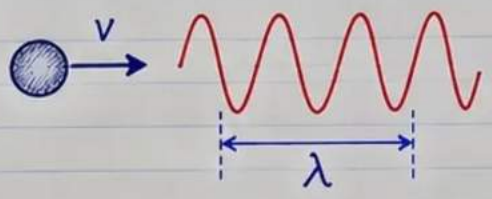
- The wavelength associated with a moving particle is given by :

$$\lambda = \frac{h}{p}$$

h = Planck's constant (6.63×10^{-34} J s)

p = momentum of particle ($p = mv$)

Wave Nature of a Moving Particle



Derivation of de Broglie Wavelength :

- Energy of a photon : $E = h\nu$
- Momentum of a photon : $p = \frac{E}{c} = \frac{h\nu}{c}$
- Since $c = \lambda\nu$, we get : $p = \frac{h}{\lambda}$
- Extending this concept to material particles, de Broglie proposed :

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Relation with Kinetic Energy :

- Kinetic energy : $K = \frac{1}{2}mv^2$
- Momentum : $p = mv = \sqrt{2mK}$
- de Broglie wavelength :

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Units :

- h : J s
- p : kg m s^{-1}
- λ : metre (m)

Important Points :

1. de Broglie wavelength is significant for microscopic particles.
2. For macroscopic bodies, λ is extremely small and cannot be observed.
3. It confirms the wave-particle duality of matter.
4. It is a fundamental concept in quantum mechanics.

Examples :

1. Electron accelerated through 100 V :

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}}\ \text{nm} \longrightarrow \text{For } V = 100\ \text{V}, \quad \lambda = \frac{1.227}{\sqrt{100}} = 0.123\ \text{nm}$$

2. A cricket ball ($m = 0.15$ kg, $v = 30$ m/s) :

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.15 \times 30} = 1.47 \times 10^{-35}\ \text{m} \rightarrow \text{Extremely small (not observable)}$$

Applications of Matter Waves :

1. Explains Davisson-Germer experiment.
2. Electron diffraction (used in electron microscope).
3. Helps in understanding atomic structure.
4. Foundation of quantum mechanics.
5. Used in modern technologies like electron microscope, nanotechnology etc.

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17. Simple Harmonic Oscillator (1-D) – Equation & Energy Levels

1. Classical Simple Harmonic Oscillator :

- A particle of mass m attached to a spring with spring constant k executes simple harmonic motion (SHM).
- The potential energy is :

$$V(x) = \frac{1}{2}kx^2$$

- Equation of motion (classical) :

$$m\frac{d^2x}{dt^2} + kx = 0$$

- Angular frequency :

$$\omega = \sqrt{\frac{k}{m}}$$

2. Time-Independent Schrödinger Equation :

- For a quantum harmonic oscillator, the potential is $V(x) = \frac{1}{2}kx^2$. The TISE is :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}kx^2\psi(x) = E\psi(x)$$

- Using angular frequency $\omega = \sqrt{\frac{m}{m}}$, this can be written as :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x)$$

- This is a second-order differential equation with discrete energy eigenvalues.
- Introduce the dimensionless variable :

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x$$

3. Solution (Eigenfunctions) :

- The normalized eigenfunctions are given by :

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{\frac{-\xi^2}{2}}$$

where,

- $H_n(\xi)$ = Hermite polynomial of order n
- $n = 0, 1, 2, 3, \ldots$
- $\xi = \sqrt{\frac{m\omega}{\hbar}}x$
- The first few eigenfunctions :

$$\psi_0(x) \propto e^{-\xi^2/2}$$
$$\psi_1(x) \propto \xi\, e^{-\xi^2/2}$$
$$\psi_2(x) \propto (2\xi^2 - 1)\, e^{-\xi^2/2}$$

4. Energy Eigenvalues :

- The allowed energy levels are :

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega \quad (n = 0, 1, 2, 3, \ldots)$$

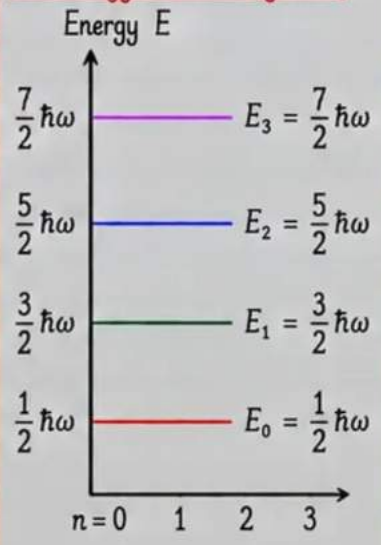
- Energy is quantized and equally spaced.
- The lowest energy (ground state) is not zero :

$$E_0 = \frac{1}{2}\hbar\omega \quad \text{(zero-point energy)}$$

- Energy difference between consecutive levels :

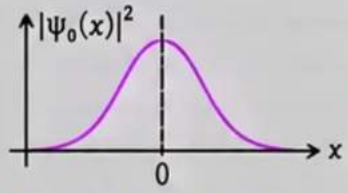
$$\Delta E = E_{n+1} - E_n = \hbar\omega$$

5. Energy Level Diagram :

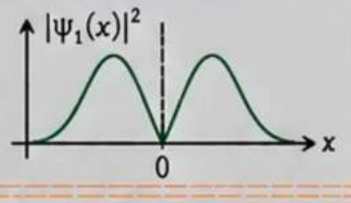


6. Probability Density :

- Probability density is $|\psi_n(x)|^2$.
- It gives the probability of finding the particle at position x.
- Ground state (n = 0) :



- First excited state (n = 1) :



7. Properties :

1. Discrete and equally spaced energy levels.
2. Non-zero ground state energy (zero-point energy).
3. Eigenfunctions involve Hermite polynomials.
4. Probability density is symmetric about x = 0.
5. Higher n ⇒ more nodes (n nodes in $\psi_n(x)$).
6. The particle is most likely found near x = 0 for ground state.
7. Classical and quantum results are consistent in the high energy limit (large n).

8. First Few Energy Levels :

n	Energy E_n
0	$\frac{1}{2}\hbar\omega$
1	$\frac{3}{2}\hbar\omega$
2	$\frac{5}{2}\hbar\omega$
3	$\frac{7}{2}\hbar\omega$

9. Summary :

- A 1-D quantum harmonic oscillator has quantized energy levels given by $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$.
- Eigenfunctions involve Hermite polynomials and are normalizable.
- The ground state energy is non-zero.
- Probability density shows n nodes for the n-th excited state.
- This model is important in many areas of physics (molecular vibrations, phonons, etc.).

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15. 1-D Box – Energy Eigenvalues, Eigenfunctions & Probability

1. Potential and Boundary Conditions :

- A particle of mass m is confined in a 1-D box of length L $(0 \le x \le L)$ with infinite potential walls.

$$V(x) = \begin{cases} 0 & (0 < x < L) \\ \infty & (x \le 0 \text{ or } x \ge L) \end{cases}$$

- Inside the box $(0 < x < L)$: $V(x) = 0$
- Outside the box : $V(x) = \infty$
- Wave function must be zero at the walls :

$$\psi(0) = 0 \quad \text{and} \quad \psi(L) = 0$$

2. Time-Independent Schrödinger Equation :

- Inside the box $(V(x) = 0)$, the TISE becomes :

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

where, $k^2 = \dfrac{2mE}{\hbar^2} \Rightarrow k = \dfrac{\sqrt{2mE}}{\hbar}$

- General solution :

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

- Apply boundary conditions :
  1. At $x = 0$: $\psi(0) = 0 \Rightarrow B = 0$
  2. At $x = L$: $\psi(L) = A\sin(kL) = 0$

For non-trivial solution $(A \neq 0)$:

$$\sin(kL) = 0 \Rightarrow kL = n\pi \quad (n = 1, 2, 3, \ldots)$$

Hence, $k = \dfrac{n\pi}{L}$

3. Energy Eigenvalues :

- Using $k = \dfrac{n\pi}{L}$ in $E = \dfrac{\hbar^2 k^2}{2m}$, we get :

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (n = 1, 2, 3, \ldots)$$

- Energy is quantized (discrete).
- Energy increases as n^2.

4. Normalized Eigenfunctions :

- With $B = 0$ and $k = \dfrac{n\pi}{L}$:

$$\psi_n(x) = A\sin\left(\frac{n\pi x}{L}\right)$$

- Normalization condition : $\displaystyle\int_0^L |\psi_n(x)|^2\,dx = 1$

$$\int_0^L A^2\sin^2\left(\frac{n\pi x}{L}\right)dx = 1$$

$$\Rightarrow A^2\left(\frac{L}{2}\right) = 1 \Rightarrow A = \sqrt{\frac{2}{L}}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right) \quad (n = 1, 2, 3, \ldots)$$

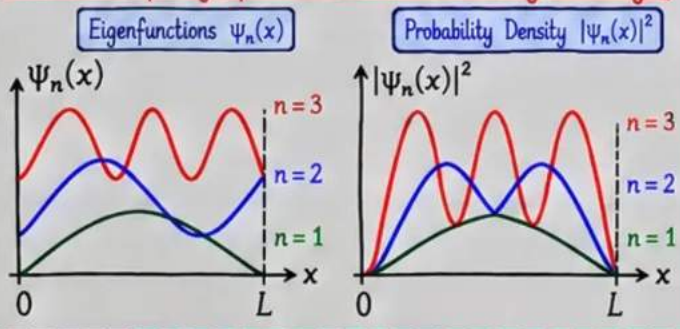
5. Probability Density :

- Probability density is :

$$|\psi_n(x)|^2 = \frac{2}{L}\sin^2\left(\frac{n\pi x}{L}\right)$$

- It gives the probability of finding the particle at position x.
- The probability is zero at the walls ($x = 0$ and $x = L$).
- Total probability : $\displaystyle\int_0^L |\psi_n(x)|^2\,dx = 1$.

6. Plots of Eigenfunctions and Probability Density :



7. Properties :

1. Energy levels are discrete : $E_n \propto n^2$.
2. $\psi_n(x)$ has $n - 1$ internal nodes.
3. $|\psi_n(x)|^2$ has n maxima.
4. Probability is zero at the walls ($x = 0$ and $x = L$).
5. Higher n $\Rightarrow$ more nodes and higher energy.
6. This is an idealized model to understand quantum confinement.

8. Summary :

- For a 1-D box (infinite potential well), energy eigenvalues are

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (n = 1, 2, 3, \ldots)$$

- Normalized eigenfunctions : $\psi_n(x) = \sqrt{\dfrac{2}{L}}\sin\left(\dfrac{n\pi x}{L}\right)$.
- Probability density : $|\psi_n(x)|^2 = \dfrac{2}{L}\sin^2\left(\dfrac{n\pi x}{L}\right)$.
- Wave function is zero at the boundaries.
- The particle has $n - 1$ internal nodes and $|\psi_n(x)|^2$ has n maxima.
- Energy is quantized and increases with n^2.

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9. Operators, Eigenvalues, Eigenfunctions & Expectation Value

1. Quantum Mechanical Operator :

- In quantum mechanics, a physical quantity is represented by a linear operator.
- Operators act on a wavefunction (Ψ) and give another wavefunction.
- Each measurable physical quantity (like position, momentum, energy etc.) has a corresponding operator.
- Operators are usually denoted by a hat (^).

$$\hat{A}\,\Psi(x) = \text{result (another function)}$$

Some Important Operators :

Physical Quantity	Operator ($\hat{A}$)
Position (x)	$\hat{x} = x$
Momentum (p)	$\hat{p} = -i\hbar \frac{d}{dx}$
Kinetic Energy (T)	$\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$
Total Energy (E)	$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$

2. Eigenvalues and Eigenfunctions :

- When an operator acts on a wavefunction and gives a constant multiple of the same wavefunction, that wavefunction is called an eigenfunction.
- The constant is called the eigenvalue.
- Mathematically :

$$\hat{A}\,\Psi_n = a_n \Psi_n$$

- Here, Ψ_n is the eigenfunction and a_n is the eigenvalue corresponding to operator $\hat{A}$.
- Only for certain allowed values (eigenvalues) we get physically acceptable solutions.

Example : Momentum Operator

$\hat{p} = -i\hbar \frac{d}{dx}$ Let $\Psi(x) = A e^{ikx}$

$$\hat{p}\,\Psi(x) = -i\hbar \frac{d}{dx}\left(A e^{ikx}\right)$$

$$= -i\hbar\,(ik)\,A e^{ikx}$$

$$= \hbar k\,\Psi(x)$$

$\Psi(x)$ is eigenfunction of momentum operator

Eigenvalue = $\hbar k$

3. Properties of Operators :

1. Linearity : $\hat{A}(a\psi_1 + b\psi_2) = a\hat{A}\psi_1 + b\hat{A}\psi_2$
2. Hermitian (Self-adjoint) : $\hat{A} = \hat{A}^\dagger$

   Hermitian operators have real eigenvalues (corresponding to measurable quantities).
3. Commutation : $[\hat{A}, \hat{B}] = AB - \hat{B}\hat{A}$

   If $[\hat{A}, \hat{B}] = 0$, operators commute and can have common eigenfunctions.
4. Orthogonality : Eigenfunctions corresponding to different eigenvalues are orthogonal.

$$\int \Psi_m^*(x)\,\Psi_n(x)\,dx = 0 \quad (\text{for } m \neq n)$$

4. Expectation Value :

- Expectation value (average value) of an observable $\hat{A}$ gives the average result when a measurement is repeated many times on identically prepared systems.
- It is given by :

$$\langle A \rangle = \int_{-\infty}^{\infty} \Psi^*(x)\,\hat{A}\,\Psi(x)\,dx$$

- Here, $\Psi(x)$ is the normalized wavefunction.
- $\langle A \rangle$ is a real number for Hermitian operator.
- Expectation value depends on the state of the system (Ψ).
- For normalized wavefunction :

$$\int_{-\infty}^{\infty} |\Psi(x)|^2\,dx = 1$$

5. Example : Expectation Value of Position

- For a normalized wavefunction $\Psi(x)$:

$$\langle x \rangle = \int_{-\infty}^{\infty} \Psi^*(x)\,x\,\Psi(x)\,dx$$

- It gives the average position of the particle.
- This value may be zero or non-zero depending on the wavefunction.

6. Important Points :

1. Operators represent observable physical quantities.
2. Eigenfunctions give possible states of the system.
3. Eigenvalues are the possible measured values.
4. Expectation value gives the average value, not a single measurement result.
5. Hermitian operators have real eigenvalues and orthogonal eigenfunctions.

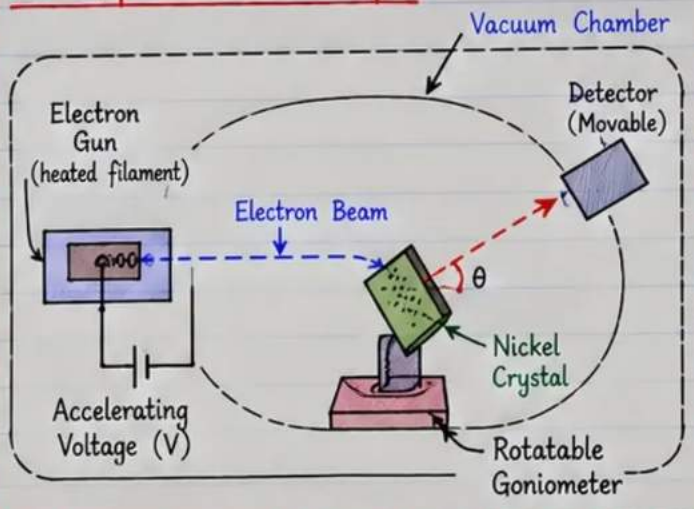
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7. Davisson-Germer Experiment

1. Introduction :

- In 1927, C. J. Davisson and L. H. Germer performed an experiment to verify the **wave nature of electrons.**
- They observed **diffraction of electrons** when they were incident on a **nickel crystal.**
- The experiment confirmed the validity of de Broglie hypothesis ($\lambda = h/p$) for matter waves.

2. Experimental Setup :



3. Key Formula :

$$\lambda = \frac{h}{\sqrt{2m_e eV}}$$

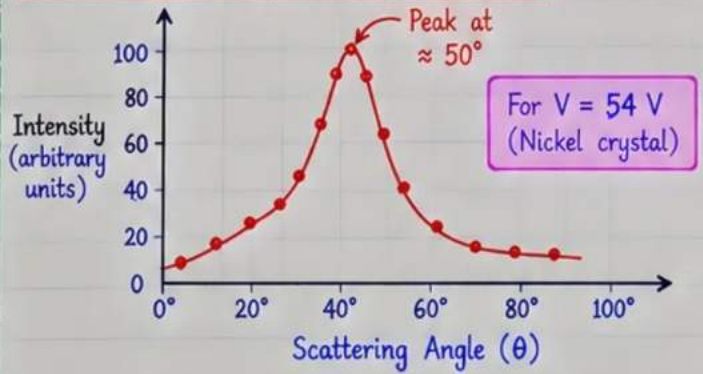
Where :

- h : Planck's constant (6.63×10^{-34} J s)
- m_e : mass of electron (9.11×10^{-31} kg)
- e : charge of electron (1.6×10^{-19} C)
- V : accelerating voltage (V)
- λ : de Broglie wavelength (m)

4. Observation :

- When the detector is rotated, the intensity of diffracted electrons is measured.
- A **sharp maximum** in intensity is observed at a particular angle ($\approx 50°$) for an accelerating voltage of about **54 V.**
- This maximum corresponds to constructive interference of electron waves reflected from different crystal planes.

5. Graph (Experimental Result) :



6. Result and Conclusion :

- The observed peak at $\theta \approx 50°$ matches the value calculated using Bragg's law.
- The calculated wavelength of electrons agrees with the de Broglie wavelength ($\lambda = h/p$).
- Hence, electrons exhibit **wave nature** and can undergo diffraction, just like light.
- This experiment provided direct experimental evidence for the existence of **matter waves.**

7. Calculation (Using Bragg's Law) :

- Bragg's law : $n\lambda = 2d \sin\theta$
- For Nickel crystal ($d \approx 0.091$ nm), $\theta \approx 50°$, $n = 1$

$$\lambda = 2d \sin\theta$$
$$= 2 \times 0.091 \times 10^{-9} \times \sin 50°$$
$$\approx 1.39 \times 10^{-10} \text{ m} \quad (0.139 \text{ nm})$$

Using de Broglie formula for V = 54 V :

$$\lambda = \frac{h}{\sqrt{2m_e eV}}$$
$$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19} \times 54}}$$
$$\approx 1.38 \times 10^{-10} \text{ m} \quad (0.138 \text{ nm})$$

8. Important Points :

1. Confirmed the wave nature of electrons.
2. Verified de Broglie hypothesis ($\lambda = \frac{h}{p}$).
3. Electrons show diffraction from crystal planes.
4. Peak intensity occurs at a definite angle.
5. Provided direct experimental evidence for matter waves.
6. Similar to X-ray diffraction, but here electrons are used.

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4. Heisenberg Uncertainty Principle

What is Heisenberg Uncertainty Principle ?

- Heisenberg Uncertainty Principle states that it is impossible to measure simultaneously the exact position and exact momentum of a particle.
- There is a fundamental limit to the accuracy with which certain pairs of physical quantities can be known.
- This is not due to experimental errors, but is a basic property of nature.

Mathematical Expression :

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} = \frac{\hbar}{2}$$

h : Planck's constant (6.63×10^{-34} Js)

$\hbar = \frac{h}{2\pi}$ (1.055×10^{-34} Js)

- Δx : Uncertainty in position (metre, m)
- Δp : Uncertainty in momentum (kg m s^{-1})

Derivation (Conceptual) :

① Consider a particle (electron) and represent it by a wave packet of wavelength range $\Delta\lambda$ (confined in length Δx).

② From wave nature : $\Delta x \approx \frac{\lambda}{2}$ (approx.)

③ Momentum of particle : $p = \frac{h}{\lambda}$

④ Uncertainty in momentum : $\Delta p \approx \frac{\hbar}{\Delta\lambda}$

⑤ Combining : $\Delta x \cdot \Delta p \approx \frac{h}{4\pi} = \frac{\hbar}{2}$

Other Forms of Uncertainty Relations :

① Position and momentum : $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$

② Energy and time : $\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$

③ Angular position and angular momentum : $\Delta\theta \cdot \Delta L \geq \frac{\hbar}{2}$

Important Points :

① Exact values of position and momentum cannot be known simultaneously.

② It is a fundamental principle of quantum mechanics.

③ It arises due to the wave nature of matter.

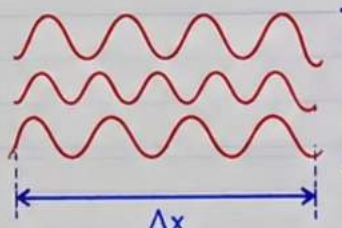
④ It is significant only for microscopic particles.

⑤ For macroscopic bodies, the uncertainty is negligibly small, so classical mechanics works.

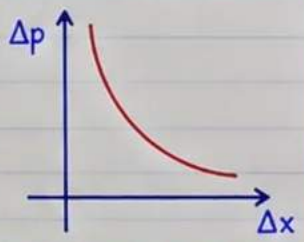
Physical Meaning :

- The exact position (x) and exact momentum (p) of a particle cannot be known at the same time.
- If we measure position more accurately ($\Delta x \downarrow$), the uncertainty in momentum ($\Delta p \uparrow$).
- Similarly, if we measure momentum more accurately ($\Delta p \downarrow$), the uncertainty in position ($\Delta x \uparrow$).
- This applies to microscopic particles (e.g., electron, proton) and not to macroscopic bodies (for them Δx and Δp are extremely small).

Wave Packet Representation :



Graphical Representation :



As Δx decreases, Δp increases and vice versa. The product $\Delta x \cdot \Delta p$ is always $\geq \hbar/2$.

Units :

- Δx : metre (m)
- Δp : kg m s^{-1}
- h : Js
- $\hbar$: Js $\left(= \frac{h}{2\pi}\right)$

Limitations :

- It does not mean we cannot measure either position or momentum separately.
- It only sets a limit on the simultaneous measurement.
- It is not due to instrument imperfections.

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18. Complete Revision – Formulas + Important Derivations + Exam Questions

1. Important Formulas :

① de Broglie wavelength :

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

② Heisenberg Uncertainty :

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}$$
$$\Delta E\,\Delta t \geq \frac{\hbar}{2}$$

③ Schrödinger Equation (T.D.) :

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$$

④ TISE (time-independent) :

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

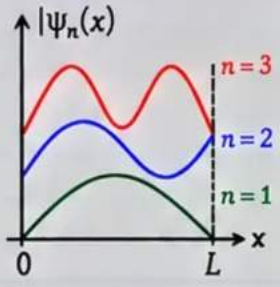
⑤ Energy of free particle :

$$E = \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m}$$

2. 1-D Infinite Potential Box :

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$



3. 3-D Box :

$$E_{n_xn_yn_z} = \frac{\hbar^2}{8m}\left(\frac{n_x^2}{L_x^2}+\frac{n_y^2}{L_y^2}+\frac{n_z^2}{L_z^2}\right)$$

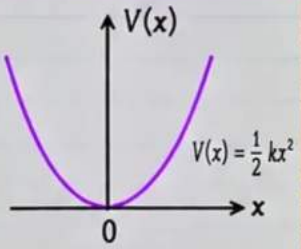
$$\psi_{n_xn_yn_z} = \sqrt{\frac{8}{L_xL_yL_z}}\sin\left(\frac{n_x\pi x}{L_x}\right) \times \sin\left(\frac{n_y\pi y}{L_y}\right)\sin\left(\frac{n_z\pi x}{L_z}\right)$$

Degeneracy (cubic) :
g = number of permutations of (n_x, n_y, n_z)
Same value of $(n_x^2+n_y^2+n_z^2)$ ⇒ same E

4. Simple Harmonic Oscillator :

$$E_n = \left(n+\frac{1}{2}\right)\hbar\omega$$

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{\frac{-\xi^2}{2}}$$
$$\xi = \sqrt{\frac{m\omega}{\hbar}}\,x$$



5. Important Derivations (Steps) :

① de Broglie Wavelength :

From $p = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{p} = \frac{h}{mv}$

② Time-Dependent Schrödinger Equation :
Using $E = i\hbar\frac{\partial}{\partial t}$ and classical Hamiltonian $E = \frac{p^2}{2m} + V$, replace $p \to -i\hbar\nabla$.
Derive : $i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$

③ Time-Independent Schrödinger Equation :
Assume $\psi(r,t) = \psi(r)\,e^{-iEt/\hbar}$
Substitute in TDSE to get :

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

④ Energy levels in 1-D Box :
Solve $\frac{d^2\psi}{dx^2} + k^2\psi = 0$ with $\psi(0) = 0,\ \psi(L) = 0$
$\Rightarrow k = \frac{n\pi}{L} \Rightarrow E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$

⑤ Simple Harmonic Oscillator :
Solve $\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{1}{2}kx^2\right)\psi = 0$
Using $\xi = \sqrt{\frac{m\omega}{\hbar}}\,x$, obtain Hermite equation
$\Rightarrow E_n = \left(n+\frac{1}{2}\right)\hbar\omega$ and $\psi_n(x)$ with $H_n(\xi)$.

6. Key Results & Concepts :

- Matter waves : $\lambda = h/p$
- Uncertainty : $\Delta x\,\Delta p \geq \frac{\hbar}{2}$
- Free particle : $E = \frac{p^2}{2m}$, $\psi \propto e^{ikx}$
- 1-D box : $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$
- 3-D box : $E_{n_xn_yn_z} = \frac{\hbar^2}{8m}\left(\frac{n_x^2}{L_x^2}+\frac{n_y^2}{L_y^2}+\frac{n_z^2}{L_z^2}\right)$
- Degeneracy : Same $(n_x^2+n_y^2+n_z^2)$
- SHO : $E_n = \left(n+\frac{1}{2}\right)\hbar\omega$
- Probability density : $|\psi(x)|^2$
- Normalization : $\int|\psi|^2\,d\tau = 1$
- Operators : $\hat{p} = -i\hbar\nabla$, $\hat{x} = x$
- Expectation value : $\langle A\rangle = \int\psi^*\hat{A}\psi\,d\tau$

7. Normalization Conditions :

- 1-D Box : $\int_0^L|\psi_n(x)|^2\,dx = 1$
- 3-D Box : $\int|\psi_{n_xn_yn_z}(x,y,z)|^2\,d\tau = 1$
- SHO : $\int_{-\infty}^{\infty}|\psi_n(x)|^2\,dx = 1$

8. Probability Density :

- Probability of finding particle between a and b :
$$P(a \leq x \leq b) = \int_a^b|\psi(x)|^2\,dx$$
- Total probability : $\int|\psi|^2\,d\tau = 1$
- Nodes : points where $\psi(x) = 0$
- 1-D box : $n-1$ internal nodes
- SHO : n nodes

9. Important Points :

① Quantization arises due to boundary conditions.
② Energy levels are discrete.
③ Wavefunction must be single-valued, finite and normalizable.
④ Degeneracy depends on symmetry (e.g., 3-D box).
⑤ Probability density gives measurable results.

10. Exam Questions (with Answers) :

Q1. What is de Broglie wavelength of an electron with velocity 2×10^6 m/s ?
Ans: $\lambda = \frac{h}{mv} \approx 3.63\times10^{-10}$ m.

Q2. State Heisenberg uncertainty principle.
Ans: $\Delta x\,\Delta p \geq \frac{\hbar}{2}$ and $\Delta E\,\Delta t \geq \frac{\hbar}{2}$.

Q3. Write the time-independent Schrödinger equation.
Ans: $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$.

Q4. Find the allowed energies for a particle in a 1-D box of length L.
Ans: $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$, $(n = 1, 2, 3, \ldots)$.

Q5. Write the normalized wavefunction for 1-D box.
Ans: $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$.

Q6. How many nodes are present for $n = 4$ in 1-D box?
Ans: $n-1 = 3$ nodes.

Q7. Write the energy eigenvalues for a particle in a 3-D cubic box ($L_x = L_y = L_z = L$).
Ans: $E_{n_xn_yn_z} = \frac{\hbar^2}{8mL^2}\left(n_x^2+n_y^2+n_z^2\right)$.

Q8. What is the degeneracy of the state (1,1,2) in a cubic box?
Ans: Degeneracy = 3 (permutations of 1, 1, 2).

Q9. Write the energy levels of a 1-D simple harmonic oscillator.
Ans: $E_n = \left(n+\frac{1}{2}\right)\hbar\omega$.

Q10. Write the ground state wavefunction of SHO.
Ans: $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\xi^2/2}$, $\xi = \sqrt{\frac{m\omega}{\hbar}}\,x$.

Q11. What is the probability density and normalization condition?
Ans: $\rho(x) = |\psi(x)|^2$, $\int|\psi|^2\,d\tau = 1$.

Q12. Why is energy quantized in a potential well?
Ans: Due to boundary conditions (wavefunction must be zero at the walls), only certain wavelengths fit inside the well.

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13. Free Particle – Schrödinger Equation & Solution

1. Free Particle :

- A free particle is a particle moving in free space with no external force acting on it.
- Hence, the potential energy is zero everywhere :

$$V(x) = 0$$

- The particle is not confined (no boundaries) and can move anywhere in space.
- It exhibits plane wave solutions.

2. Time-Independent Schrödinger Equation :

- For a free particle ($V(x) = 0$), the TISE becomes :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

- This is a second-order linear differential equation with constant coefficients.
- Here, $E > 0$ (total energy of the free particle).

3. Solution of the Equation :

- Rearrange the equation :

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

where, $k^2 = \frac{2mE}{\hbar^2} \Rightarrow k = \frac{\sqrt{2mE}}{\hbar}$

- The general solution is :

$$\psi(x) = A\,e^{ikx} + B\,e^{-ikx}$$

- This represents a superposition of two plane waves : one moving in +x direction and one in −x direction.
- Here, A and B are constants (complex in general).

4. Plane Wave Solution :

- A plane wave solution of the free particle is :

$$\psi_k(x) = A\,e^{ikx} \quad \text{where,} \quad k = \frac{\sqrt{2mE}}{\hbar}$$

- This is a traveling wave with definite momentum.
- Its complex conjugate is :

$$\psi_k^*(x) = A^*\,e^{-ikx}$$

- Both e^{ikx} and e^{-ikx} are equally valid solutions.

5. Energy and Momentum :

- The energy of the free particle is :

$$E = \frac{\hbar^2k^2}{2m} = \frac{p^2}{2m}$$

- Its momentum is : $p = \hbar k$
- Thus, the plane wave $\psi_k(x) = e^{ikx}$ represents a particle with definite momentum $p = \hbar k$ and definite energy $E = \frac{p^2}{2m}$.

6. Physical Interpretation :

1. The plane wave e^{ikx} is an eigenfunction of the momentum operator $\hat{p} = -i\hbar\frac{d}{dx}$:

$$\hat{p}\psi_k(x) = p\psi_k(x) = \hbar k\,\psi_k(x)$$

2. Hence, a free particle has a definite momentum.
3. However, plane waves are not normalizable (over $-\infty$ to $+\infty$), so they are not physically acceptable as true wave functions.
4. In practice, we use a wave packet (superposition of plane waves) to represent a real free particle.

7. Probability Density :

- For the plane wave $\psi_k(x) = A\,e^{ikx}$,

$$|\psi_k(x)|^2 = |A|^2 \text{ (constant)}$$

- The probability density is uniform in space, which means the particle can be found everywhere with equal probability.
- This reflects the free and unconfined nature of the particle.

8. Summary :

- For a free particle, $V(x) = 0$.
- The TISE reduces to $\frac{d^2\psi}{dx^2} + k^2\psi = 0$.
- Solution : $\psi(x) = A\,e^{ikx} + B\,e^{-ikx}$.
- $E = \frac{\hbar^2k^2}{2m}$, $p = \hbar k$.
- Plane wave is an eigenfunction of momentum operator.
- Probability density is constant (not normalizable).
- A real free particle is described by a wave packet.

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16. Particle in 3-D Box – Energy Levels & Degeneracy

1. 3-D Infinite Potential Box :

- A particle of mass m is confined in a rectangular box with sides L_x, L_y, L_z and infinite potential walls.
- $V(x,y,z) = 0$ inside the box
- $V(x,y,z) = \infty$ outside the box

$$V(x,y,z) = \begin{cases} 0 & (0<x<L_x, 0<y<L_y, 0<z<L_z) \\ \infty & (\text{otherwise}) \end{cases}$$

2. Time-Independent Schrödinger Equation :

- Inside the box, the TISE is :

$$-\frac{\hbar^2}{2m}\nabla^2\psi(x,y,z) = E\psi(x,y,z)$$

- Since $V = 0$ inside, it becomes :

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + k^2\psi = 0$$

where, $k^2 = \frac{2mE}{\hbar^2}$

3. Separation of Variables :

- Assume : $\psi(x,y,z) = X(x)\,Y(y)\,Z(z)$
- Substituting and separating variables gives :

$$\frac{d^2X}{dx^2} + k_x^2 X = 0 \quad \Big| \quad \frac{d^2Y}{dy^2} + k_y^2 Y = 0 \quad \Big| \quad \frac{d^2Z}{dz^2} + k_z^2 Z = 0$$

where, $k^2 = k_x^2 + k_y^2 + k_z^2$

4. Boundary Conditions :

- Since potential is infinite at the walls :

$$\psi(0,y,z) = \psi(L_x,y,z) = 0$$
$$\psi(x,0,z) = \psi(x,L_y,z) = 0$$
$$\psi(x,y,0) = \psi(x,y,L_z) = 0$$

- Non-trivial solutions require :

$$k_x = \frac{n_x\pi}{L_x}, \quad k_y = \frac{n_y\pi}{L_y}, \quad k_z = \frac{n_z\pi}{L_z}$$

$n_x, n_y, n_z = 1, 2, 3, \ldots$

5. Eigenfunctions (Normalized) :

- The normalized wave function is :

$$\psi_{n_x n_y n_z}(x,y,z) = \sqrt{\frac{8}{L_x L_y L_z}} \sin\left(\frac{n_x\pi x}{L_x}\right)\sin\left(\frac{n_y\pi y}{L_y}\right)\sin\left(\frac{n_z\pi z}{L_z}\right)$$

$n_x, n_y, n_z = 1, 2, 3, \ldots$

6. Energy Eigenvalues :

- The energy of the particle is :

$$E_{n_x n_y n_z} = \frac{\hbar^2}{8m}\left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right)$$

where, $n_x, n_y, n_z = 1, 2, 3, \ldots$
(h : Planck's constant)

7. Degeneracy :

- For a cubic box ($L_x = L_y = L_z = L$), energy depends only on $n_x^2 + n_y^2 + n_z^2$:

$$E_{n_x n_y n_z} = \frac{\hbar^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

- Different combinations of (n_x, n_y, n_z) having the same value of $(n_x^2+n_y^2+n_z^2)$ give the same energy ⇒ degenerate states.
- Degeneracy arises due to the symmetry of the cubic box.
- General rule (for distinct non-zero n_x, n_y, n_z) :

- All different : degeneracy = 6 (permutations of n_x, n_y, n_z)
- Two equal : degeneracy = 3 (e.g., (1,1,2))
- All equal : degeneracy = 1 (e.g., (1,1,1))

8. Examples (Cubic Box : $L_x = L_y = L_z = L$) :

(n_x, n_y, n_z)	$n_x^2+n_y^2+n_z^2$	Energy E	Degeneracy
(1,1,1)	3	$3\frac{\hbar^2}{8mL^2}$	1
(1,1,2)	6	$6\frac{\hbar^2}{8mL^2}$	3
(1,2,2)	9	$9\frac{\hbar^2}{8mL^2}$	3
(1,2,3)	14	$14\frac{\hbar^2}{8mL^2}$	6
(2,2,2)	12	$12\frac{\hbar^2}{8mL^2}$	1

9. Key Points :

1. Energy levels are quantized and depend on $n_x^2+n_y^2+n_z^2$.
2. Degeneracy occurs for a cubic box due to symmetry.
3. Number of degenerate states depends on how many quantum numbers are equal.
4. For a rectangular box ($L_x \neq L_y \neq L_z$), degeneracy is generally removed.
5. 3-D box is a simple model to understand quantization in three dimensions.

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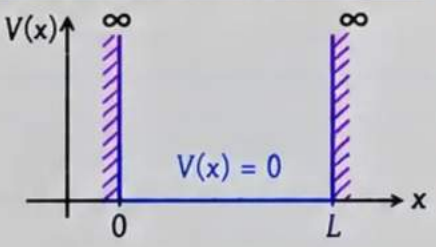
14. Particle in 1-D Box – Equation, Boundary Conditions & Solution

1. Introduction :

- A particle in a 1-D box is a particle confined in a region $0 \le x \le L$ with infinite potential walls.
- Inside the box, potential $V(x) = 0$.
- Outside the box, potential $V(x) = \infty$.
- The particle cannot exist outside the box (wave function becomes zero).

2. Potential V(x) :

$$V(x) = \begin{cases} 0 & (0 < x < L) \\ \infty & (x \le 0 \text{ or } x \ge L) \end{cases}$$



3. Time-Independent Schrödinger Equation :

- Inside the box ($0 < x < L$), $V(x) = 0$, so TISE becomes :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

or $$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

where, $k^2 = \frac{2mE}{\hbar^2}$

4. General Solution (Inside the Box) :

- The general solution of the differential equation $\frac{d^2\psi}{dx^2} + k^2\psi = 0$ is :

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

- where, $k = \frac{\sqrt{2mE}}{\hbar}$

5. Boundary Conditions :

- Since the potential is infinite outside the box, the wave function must be zero at the walls.

$$\psi(0) = 0 \quad \text{and} \quad \psi(L) = 0$$

- At $x = 0$: $\psi(0) = A\sin(0) + B\cos(0)$
  $= B = 0 \Rightarrow B = 0$
- So, $\psi(x) = A\sin(kx)$
- At $x = L$: $\psi(L) = A\sin(kL) = 0$
  For non-trivial solution ($A \neq 0$),
  $\sin(kL) = 0 \Rightarrow kL = n\pi \quad (n = 1, 2, 3, \ldots)$

$$k = \frac{n\pi}{L}$$

6. Allowed Energy Levels :

- Using $k = \frac{n\pi}{L}$ in $E = \frac{\hbar^2k^2}{2m}$:

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \qquad (n = 1, 2, 3, \ldots)$$

- Energy is quantized.
- Only discrete energy values are allowed.
- Ground state energy : $E_1 = \frac{\pi^2\hbar^2}{2mL^2}$

7. Normalized Wave Function :

- The normalized wave function is :

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right) \quad (n = 1, 2, 3, \ldots)$$

- It satisfies the normalization condition :

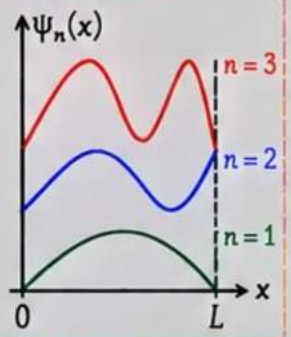
$$\int_0^L |\psi_n(x)|^2\,dx = 1$$

- Here, $A = \sqrt{\frac{2}{L}}$ (normalization constant).

8. Energy Eigenfunctions & Eigenvalues:

- Eigenfunctions : $\psi_n(x) =$ $\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$
- Eigenvalues : E_n को n से व्यक्त करें :
  $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$
- $n = 1, 2, 3, \ldots$ (quantum number)

9. Wavefunctions for First Three States :



10. Probability Density :

- Probability density for n-th state :

$$|\psi_n(x)|^2 = \frac{2}{L}\sin^2\left(\frac{n\pi x}{L}\right)$$

- It gives the probability of finding the particle at position x.
- The probability density is zero at the walls ($x = 0$ and $x = L$).
- It has n maxima inside the box.

11. Important Points :

1. Energy is quantized : $E_n \propto n^2$.
2. Wave function is zero at the walls.
3. Only certain wavelengths fit in the box.
4. Higher $n \Rightarrow$ more nodes and higher energy.
5. This is an idealized model to understand quantum confinement.

12. Summary :

- Inside the box ($0 < x < L$), $V(x) = 0$ and TISE is solved.
- Boundary conditions : $\psi(0) = 0$, $\psi(L) = 0$.
- Allowed wavefunctions : $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$.
- Allowed energy levels : $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$
- Energy is discrete and the particle has $n - 1$ internal nodes.

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11. Schrödinger Equation – Time-Dependent Equation + Derivation

1. Introduction :

- Schrödinger equation is the fundamental equation of non-relativistic quantum mechanics.
- It describes the time evolution of the wave function $\psi(x,t)$ of a quantum particle.
- It gives a relation between the total energy of the particle and its wave function.
- It is obtained by applying de Broglie hypothesis and the correspondence principle to classical energy equation.

2. Classical Energy Equation :

- For a particle of mass m moving in a potential $V(x,t)$, the total energy is :

$$E = K + V = \frac{p^2}{2m} + V(x,t)$$

- Using classical mechanics, energy and momentum are given by :

$$E \to i\hbar \frac{\partial}{\partial t} \quad \text{and} \quad p \to -i\hbar \frac{\partial}{\partial x}$$

3. Derivation of Time-Dependent Schrödinger Equation :

① Start with classical energy equation :

$$E = \frac{p^2}{2m} + V(x,t)$$

② Apply operator form ($E \to i\hbar \frac{\partial}{\partial t}$, $p \to -i\hbar \frac{\partial}{\partial x}$):

$$i\hbar \frac{\partial}{\partial t} = \left(-i\hbar \frac{\partial}{\partial x}\right)^2 + V(x,t)$$

③ Simplify $(-i)^2 = -1$:

$$i\hbar \frac{\partial}{\partial t} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x,t)$$

④ Operate on wave function $\psi(x,t)$:

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t)$$

This is the Time-Dependent Schrödinger Equation (TDSE) for a particle in one dimension.

4. Time-Dependent Schrödinger Equation :

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t)$$

- $\psi(x,t)$: wave function of the particle
- i : imaginary unit ($i^2 = -1$)
- $\hbar$: reduced Planck's constant ($h/2\pi$)
- m : mass of the particle
- $V(x,t)$: potential energy

5. General Form in 3 Dimensions :

$$i\hbar \frac{\partial \psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r},t) + V(\vec{r},t)\psi(\vec{r},t)$$

- $\vec{r} = (x, y, z)$: position vector
- ∇^2 : Laplacian operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

6. Physical Meaning :

① Describes how the wave function $\psi(x,t)$ changes with time.

② It is a linear, first-order partial differential equation in time and second-order in space.

③ It determines the allowed quantum states and their time evolution.

④ The square of the wave function $|\psi(x,t)|^2$ gives the probability density of finding the particle.

⑤ It is the central equation of non-relativistic quantum mechanics.

7. Important Points :

① It governs the time evolution of quantum systems.
② It is derived using de Broglie relations and operator method.
③ It is valid for non-relativistic particles.
④ For time-independent potential ($V = V(x)$), we can separate variables : $\psi(x,t) = \psi(x)\,T(t)$
⑤ Leads to the time-independent Schrödinger equation with energy eigenvalues.

8. Summary :

- Schrödinger equation gives the complete description of a quantum particle.
- It is derived from classical energy equation using operator approach.
- It relates the wave function $\psi(x,t)$, its spatial and temporal variations, and the potential energy.

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10. Wave Function (ψ) – Significance, Probability & Normalization

1. Introduction to Wave Function (ψ) :

- In quantum mechanics, the state of a particle is completely described by a **wave function**, represented by ψ (psi).
- ψ is a complex-valued function of position and time : $\psi(x,t)$.
- It gives complete information about the quantum state of a system.

> $\psi(x,t)$ is not a physical observable, but it is a mathematical tool to describe a quantum system.

2. Significance of Wave Function :

- ψ contains all the information about the system.
- It determines the probability of finding the particle at a particular position.
- A valid wave function must be :
  1. Single-valued (only one value at a point)
  2. Finite (cannot be infinite)
  3. Continuous (no sudden break)
  4. Square-integrable (normalizable)

3. Probability Interpretation (Born's Rule) :

- The wave function itself does not give the probability of finding a particle.
- The probability density is given by the square of the magnitude of the wave function :

$$P(x,t) = |\psi(x,t)|^2 = \psi^*(x,t)\,\psi(x,t)$$

- $P(x,t)$ is always real and non-negative.
- $P(x,t)\,dx$ gives the probability of finding the particle between x and $x + dx$.
- The total probability over all space must be 1.

4. Normalization of Wave Function :

- For a valid physical wave function, the total probability of finding the particle anywhere in space must be 1.

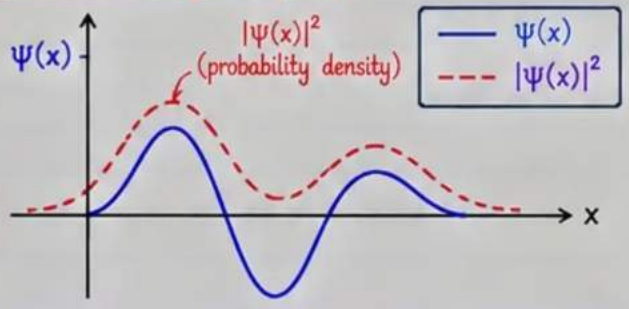
$$\int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx = 1$$

- If $\int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx \neq 1$, we normalize it.
- Normalized wave function :

$$\psi_N(x,t) = \frac{1}{\sqrt{N}}\,\psi(x,t)$$

where, $N = \int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx$

5. Probability Graph :



> The wave function $\psi(x)$ can be positive or negative, but the probability density $|\psi(x)|^2$ is always ≥ 0.

6. Example : Normalization of a Wave Function

Consider a wave function : $\psi(x) = A\,e^{-\alpha x^2}$ $(\alpha > 0)$

Normalize it :

1. $\int_{-\infty}^{\infty} |\psi(x)|^2\,dx = 1$
2. $\int_{-\infty}^{\infty} |A|^2\,e^{-2\alpha x^2}\,dx = 1$
3. $|A|^2 \sqrt{\frac{\pi}{2\alpha}} = 1 \Rightarrow A = \left(\frac{2\alpha}{\pi}\right)^{1/4}$

Normalized wave function : $\psi(x) = \left(\frac{2\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2}$

7. Key Points :

1. $|\psi(x,t)|^2$ gives probability density.
2. Total probability must be 1 (normalization).
3. ψ can be complex, but $|\psi|^2$ is real.
4. Wave function must be single-valued, finite, continuous and square-integrable.
5. Normalization ensures physical acceptability of the wave function.

8. Summary :

- Wave function (ψ) completely describes the quantum state.
- Its square modulus gives the probability density.
- It must be normalized so that total probability is 1.
- A valid wave function is single-valued, finite, continuous and square-integrable.
- Normalization constant is determined from $\int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx = 1$.

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12. Time-Independent Schrödinger Equation + Physical Significance

1. Idea (Separation of Variables):

- For a time-independent potential $V(x)$, the total wave function can be written as a product of spatial and time parts :

$$\Psi(x,t) = \psi(x)T(t)$$

- Substituting this in the time-dependent Schrödinger equation and separating variables gives two independent equations, one for space and one for time.

2. Time-Independent Schrödinger Equation :

- For a particle of mass m in a potential $V(x)$, the time-independent Schrödinger equation (TISE) is :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

- $\psi(x)$: spatial wave function (eigenfunction)
- E : total energy of the particle (eigenvalue)
- $V(x)$: potential energy (independent of time)
- $\hbar$: reduced Planck's constant ($h / 2\pi$)
- m : mass of the particle

3. Derivation (From Time-Dependent Equation) :

① Start with TDSE :

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x)\Psi(x,t)$$

② Assume : $\Psi(x,t) = \psi(x)T(t)$

③ Substitute and divide by $\psi(x)T(t)$:

$$\frac{1}{T}\frac{dT}{dt} = -\frac{\hbar^2}{2m}\frac{1}{\psi}\frac{d^2\psi}{dx^2} + V(x) = \text{constant} = -\frac{iE}{\hbar}$$

④ This gives two equations :

- Time part : $\frac{dT}{dt} = -\frac{iE}{\hbar}T \Rightarrow T(t) = e^{-iEt/\hbar}$
- Space part (TISE) :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

4. Physical Significance :

① **Quantization of Energy :**
- Non-trivial solutions ($\psi(x) \neq 0$) exist only for certain discrete values of E, called eigenvalues (allowed energies).
- For bound systems, energy levels are quantized.

② **Stationary States :**
- The spatial part $\psi(x)$ does not depend on time.
- The probability density $|\psi(x)|^2$ is time-independent for a stationary state.

③ **Eigenfunctions :**
- $\psi(x)$ are eigenfunctions of the Hamiltonian operator $\hat{H}$ (energy operator).
- Each eigenfunction corresponds to a definite energy eigenvalue E.

④ **Probability Interpretation :**
- $|\psi(x)|^2$ gives the probability density of finding the particle at position x.
- Normalization ensures total probability is 1.

5. Example : Particle in a 1D Infinite Potential Well

Potential :

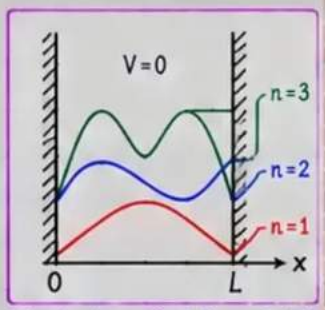
$$V(x) = \begin{cases} 0 & (0 < x < L) \\ \infty & (x \leq 0 \text{ or } x \geq L) \end{cases}$$

TISE inside the well ($V = 0$) :

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \Rightarrow \frac{d^2\psi}{dx^2} + k^2\psi = 0$$

where, $k = \sqrt{\frac{2mE}{\hbar^2}}$

Solution : $\psi(x) = A\sin(kx) + B\cos(kx)$



Wavefunctions (first three states)

Boundary conditions : $\psi(0) = 0, \quad \psi(L) = 0 \Rightarrow A = 0,$

$$\sin(kL) = 0 \Rightarrow k = \frac{n\pi}{L} \quad (n = 1, 2, 3, \ldots)$$

Normalized wave function : $\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$

Energy eigenvalues : $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (n = 1, 2, 3, \ldots)$

6. Important Points :

① Time-independent potential $V(x)$ leads to TISE.
② Energy eigenvalues are quantized for bound systems.
③ $\psi(x)$ are stationary states with definite energy.
④ $|\psi(x)|^2$ gives position probability density.
⑤ TISE is a second-order differential equation and its solutions depend on boundary conditions.

7. Summary :

- The Time-Independent Schrödinger Equation is obtained by separating variables in TDSE for time-independent potentials.
- Its solutions give stationary states (eigenfunctions) and discrete energy levels (eigenvalues).
- It fully describes the spatial behavior and allowed energies of a quantum system.

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1. Quantum Mechanics – Introduction & Dual Nature of Matter and Radiation

What is Quantum Mechanics ?

- Quantum Mechanics is the branch of physics which deals with the behaviour of matter and radiation at microscopic (atomic and subatomic) level.
- It explains the wave-particle duality and various phenomena which cannot be explained by classical physics.
- It is based on the concept of quantization of energy, uncertainty principle and wave function.

Need of Quantum Mechanics :

1. Classical physics (Newton's mechanics and Maxwell's theory) failed to explain certain experimental results.
2. Black body radiation – Ultraviolet catastrophe.
3. Photoelectric effect – Could not be explained by wave theory.
4. Atomic stability – According to classical theory, atoms should collapse.
5. Line spectra – Classical theory could not explain discrete spectral lines.

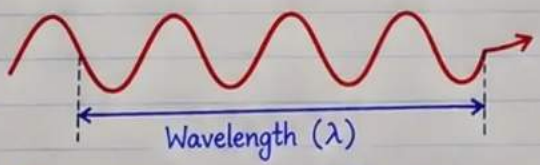
Classical vs Quantum Physics :

Aspect	Classical Physics	Quantum Mechanics
Nature of matter	Considered as particles	Shows wave-particle duality
Energy	Continuous	Quantized (discrete)
Description	Deterministic	Probabilistic
Where it works	Macroscopic scale	Microscopic (atomic) scale
Examples	Newton's laws, Maxwell's theory	Planck, Einstein, de Broglie, Schrödinger

Dual Nature of Radiation (Light) :

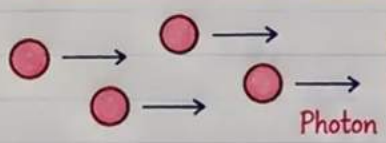
- Light behaves both as a wave and as a particle.
- Wave nature explains phenomena like interference, diffraction and polarization.
- Particle nature explains photoelectric effect, Compton effect etc.
- Thus, light has dual nature – it behaves as electromagnetic wave as well as a stream of particles called photons.

Wave Nature of Light



- Shows interference, diffraction and polarization.
- Obeys Maxwell's equations.
- Wavelength (λ) and Frequency (ν) related by : $c = \lambda\nu$

Particle Nature of Light (Photon)



$$E = h\nu = \frac{hc}{\lambda}$$

- Light consists of packets of energy called photons.
- Each photon has energy $E = h\nu$.
- Photon travels with speed $c = 3 \times 10^8$ m/s.
- Explains photoelectric effect and Compton effect.

Light has dual nature – wave in propagation and particle in interaction.

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6. Wave Packet Concept + Phase & Group Velocity

1. Wave Packet Concept :

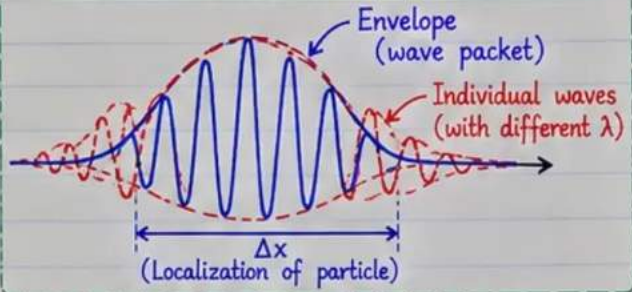
- A single wave (sinusoidal wave) has infinite extent and cannot represent a real particle.
- A real particle is represented by a group of waves (superposition of many waves with slightly different wavelengths and frequencies).
- This group of waves is called a **wave packet.**
- Wave packet is localized in space and can describe the position of a particle within a certain region (Δx).

> Wave packet = Superposition of many waves with nearby wavelengths.

Formation of Wave Packet :

- A wave packet is formed by superposition of waves with different wavelengths (λ) and frequencies (ν) around a central value.

$$\Psi(x,t) = \int_{k_0-\Delta k}^{k_0+\Delta k} A(k)\, e^{i(kx-\omega t)}\, dk$$

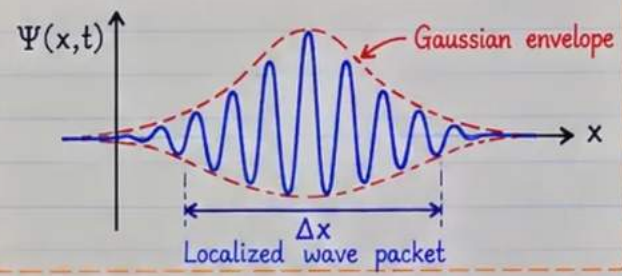


2. Mathematical Representation :

- Consider a group of waves with wave numbers around k_0 : $k = k_0 + \Delta k$
- Each wave has the form : $\Psi_k = A\, e^{i(kx-\omega t)}$
- Superposition gives a wave packet :

$$\Psi(x,t) = \int_{k_0-\Delta k}^{k_0+\Delta k} A(k)\, e^{i(kx-\omega t)}\, dk$$

3. Shape of a Wave Packet :



4. Phase Velocity (v_p) :

- Phase velocity is the velocity with which a single wave (constant phase) propagates.
- It is given by :

$$v_p = \frac{\omega}{k}$$

- It represents the speed of individual wave crests.
- For de Broglie waves (non-relativistic) :

$$\omega = \frac{E}{\hbar},\quad k = \frac{p}{\hbar} \rightarrow v_p = \frac{\omega}{k} = \frac{E}{p}$$

5. Group Velocity (v_g) :

- Group velocity is the velocity with which the envelope of the wave packet (and hence the particle) moves.
- It is given by :

$$v_g = \frac{d\omega}{dk}$$

- It represents the speed of the particle or the information.
- For a free particle : $\omega = \frac{\hbar k^2}{2m}$

$$v_g = \frac{d\omega}{dk} = \frac{d}{dk}\left(\frac{\hbar k^2}{2m}\right) = \frac{\hbar k}{m} = \frac{p}{m}$$

$v_g = \frac{p}{m}$ (= particle velocity)

6. Comparison :

Aspect	Phase Velocity (v_p)	Group Velocity (v_g)
Definition	Speed of individual wave crest	Speed of envelope (wave packet)
Expression	$v_p = \frac{\omega}{k}$	$v_g = \frac{d\omega}{dk}$
Physical meaning	Wave propagation	Particle / information propagation
For free particle	$v_p = \frac{E}{p}$	$v_g = \frac{p}{m}$
Relation	$v_p \cdot v_g = c^2$ (for relativistic particle)	

7. Important Points :

1. A real particle is represented by a wave packet, not a single wave.
2. Wave packet allows localization of the particle within a region Δx.
3. Phase velocity can be greater than the speed of light, but it does not carry energy or information.
4. Group velocity represents the actual particle velocity and always less than or equal to c.
5. Wave packet concept helps in understanding the dual nature of matter (wave + particle).

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3. Properties & Applications of Matter Waves

Properties of Matter Waves :

① **Associated with all moving material particles :**
- Every moving particle (electron, proton, neutron, atom, etc.) has a wave nature.

② **Wavelength (λ) :**
- Given by de Broglie relation $\lambda = \frac{h}{p} = \frac{h}{mv}$.

③ **Frequency (ν) :**
- Given by $\nu = \frac{E}{h}$ (E = total energy).

④ **Not electromagnetic waves :**
- These waves are not due to oscillating charges, so they do not require a medium.

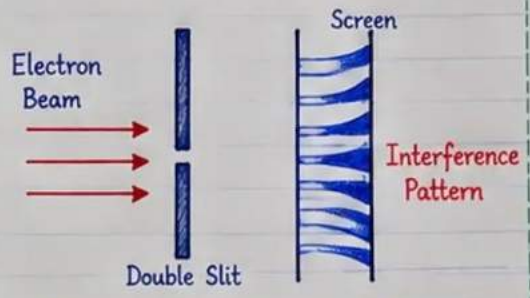
⑤ **Obey superposition principle :**
- They can interfere and diffract like ordinary waves.

⑥ **Existence significant for microscopic particles :**
- For macroscopic bodies, wavelength is extremely small and cannot be observed.

⑦ **Phase velocity and group velocity :**
- Phase velocity $v_p = \frac{c^2}{v}$ (can be greater than c).
- Group velocity $v_g = v$ (equal to particle velocity).

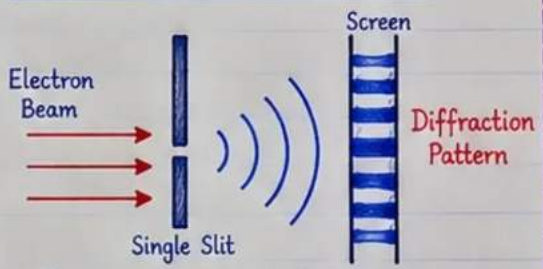
⑧ **Wave packet nature :**
- A particle is represented by a wave packet, not a single infinite wave, to satisfy the uncertainty principle.

Interference of Matter Waves



- Electron beam shows interference pattern similar to light.
- Confirms the wave nature of material particles.

Diffraction of Matter Waves



- Electron beam spreads after passing through a narrow slit.
- Confirms the wave nature of particles.

Particle Nature vs Wave Nature :

Particle Nature (Classical)	Wave Nature (de Broglie)
Well defined position and momentum	Spread over space, probability distribution
Follows Newton's laws	Shows interference and diffraction
No wavelength	Associated wavelength $\lambda = \frac{h}{p}$
No superposition	Obeys superposition principle
Classical mechanics	Quantum mechanics

Phase and Group Velocity :

- **Phase velocity :**

$$v_p = \frac{\omega}{k} = \frac{E}{p} = \frac{c^2}{v}$$

Can be greater than speed of light. (Does not carry energy)

- **Group velocity :**

$$v_g = \frac{d\omega}{dk} = v$$

Equal to particle velocity. (Carries energy and information)

Important Formulas :

① de Broglie wavelength : $\lambda = \frac{h}{p} = \frac{h}{mv}$

② Frequency : $\nu = \frac{E}{h}$

③ Phase velocity : $v_p = \frac{c^2}{v}$

④ Group velocity : $v_g = v$

⑤ Kinetic energy : $K = \frac{1}{2}mv^2$

⑥ Total energy : $E = mc^2$ (relativistic)

⑦ Momentum : $p = mv = \sqrt{2mK}$

⑧ Relation : $\lambda = \frac{h}{\sqrt{2mK}}$

Applications of Matter Waves :

① Davisson-Germer experiment (verification of matter waves)
② Electron diffraction (electron microscope)
③ Study of atomic structure
④ Neutron diffraction (crystal structure analysis)
⑤ Modern technologies (electron microscope, nanotechnology, SEM, TEM etc.)

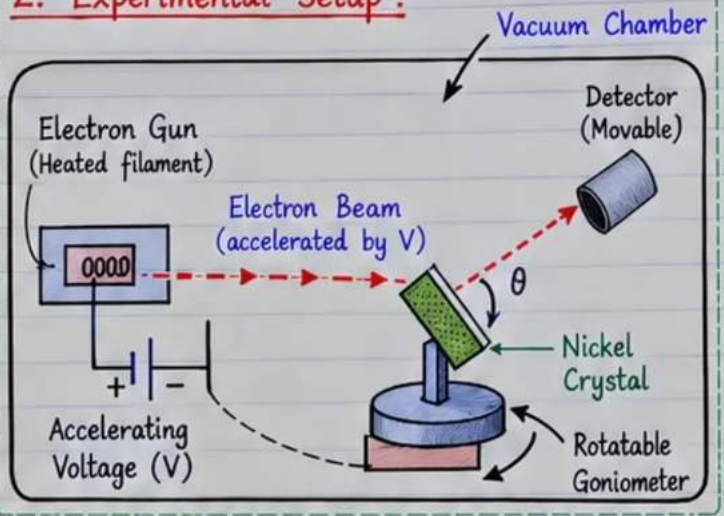
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8. Davisson - Germer Experiment

1. Introduction :

- In 1927, C. J. Davisson and L. H. Germer performed an experiment to verify the **wave nature of electrons.**
- They observed diffraction of electrons when they were incident on a **nickel crystal.**
- This experiment provided direct experimental evidence for the validity of de Broglie hypothesis ($\lambda = h/p$) for matter waves.

2. Experimental Setup :



3. Key Formulas :

de Broglie Wavelength :

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2 m_e eV}}$$

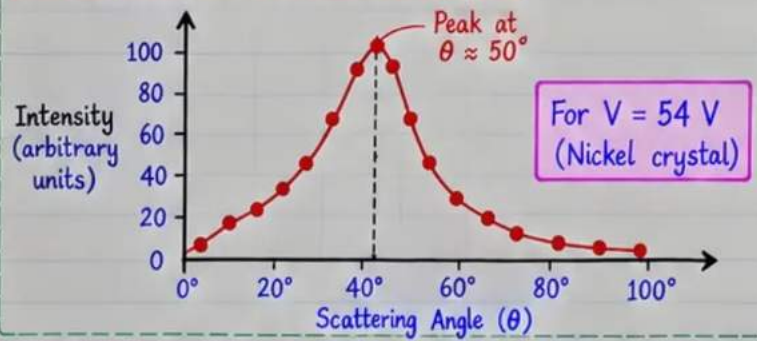
For electrons accelerated through potential V :

- h : Planck's constant (6.63×10^{-34} J s)
- m_e : mass of electron (9.11×10^{-31} kg)
- e : charge of electron (1.6×10^{-19} C)
- V : accelerating voltage (V)
- λ : de Broglie wavelength (m)

4. Observation :

- When the detector is rotated, the intensity of scattered electrons is measured.
- A **sharp maximum** in intensity is observed at a scattering angle $\theta \approx 50°$ for an accelerating voltage of about **54 V.**
- This indicates constructive interference due to diffraction of electrons from the crystal planes of nickel.
- Similar maxima are observed for other angles, satisfying Bragg's law.

5. Experimental Result (Graph) :



6. Result :

- The observed peak at $\theta \approx 50°$ matches the value calculated using Bragg's law.
- The calculated wavelength of electrons agrees with the de Broglie wavelength ($\lambda = h/p$).
- Hence, the experiment confirms the **wave nature of electrons.**

7. Calculation (Using Bragg's Law) :

- Bragg's law : $n\lambda = 2d \sin\theta$
- For nickel crystal ($d = 0.091$ nm), $\theta \approx 50°$, $n = 1$

$$\lambda = 2d \sin\theta$$
$$= 2 \times 0.091 \times 10^{-9} \times \sin 50°$$
$$\approx 1.39 \times 10^{-10} \text{ m} \quad (0.139 \text{ nm})$$

Using de Broglie formula (for V = 54 V) :

$$\lambda = \frac{h}{\sqrt{2 m_e eV}}$$
$$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.6 \times 10^{-19} \times 54}}$$
$$\approx 1.38 \times 10^{-10} \text{ m} \quad (0.138 \text{ nm})$$

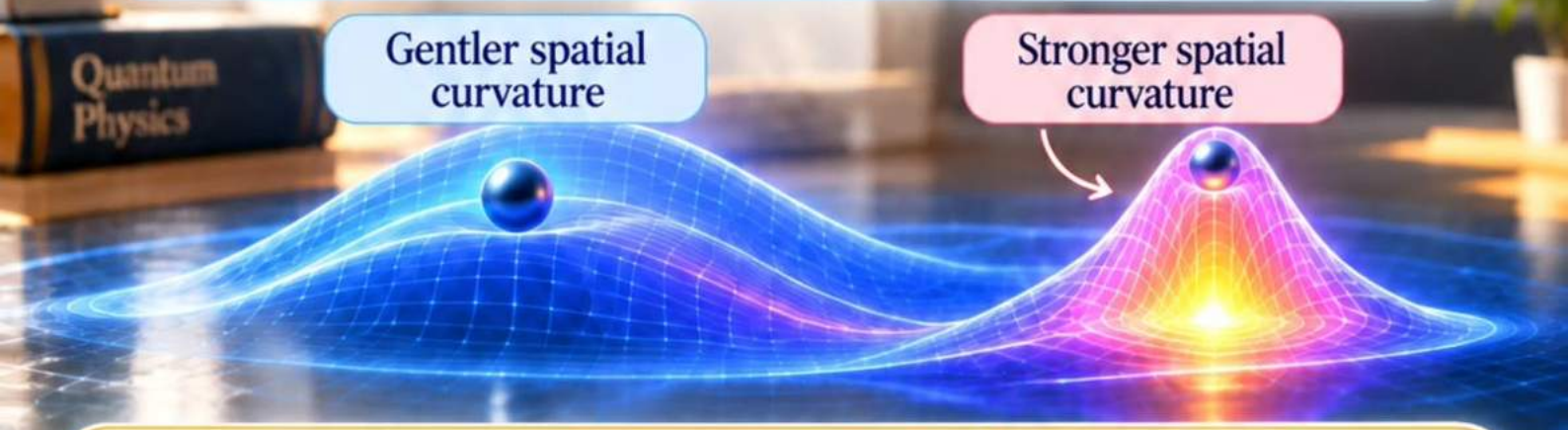
8. Conclusion :

- Electrons exhibit diffraction just like light.
- The experiment verified the de Broglie hypothesis ($\lambda = h/p$) for matter waves.
- This provided direct experimental evidence for the wave nature of electrons and confirmed the dual nature of matter (particle + wave).

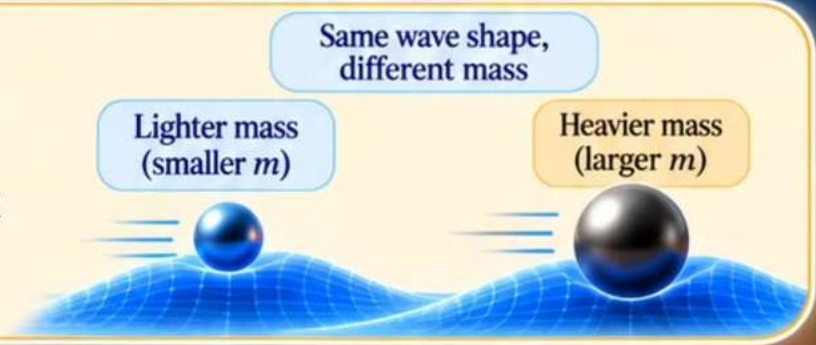
$-\hbar^2/(2m)\nabla^2$: shape, curvature, and mass

$$-\frac{\hbar^2}{2m}\nabla^2$$

1. This term contains the **spatial-curvature operator** on the wavefunction.
2. ∇^2 looks at how the wavefunction **bends in space.**
3. Together with $\hbar^2/2m$, it forms the **kinetic-energy operator.**
4. For the same wave shape, a **larger mass** m makes this term **smaller.**



Analogy: a ripple with more small-scale bending has **more structure,** and a heavier object is **harder to accelerate.**



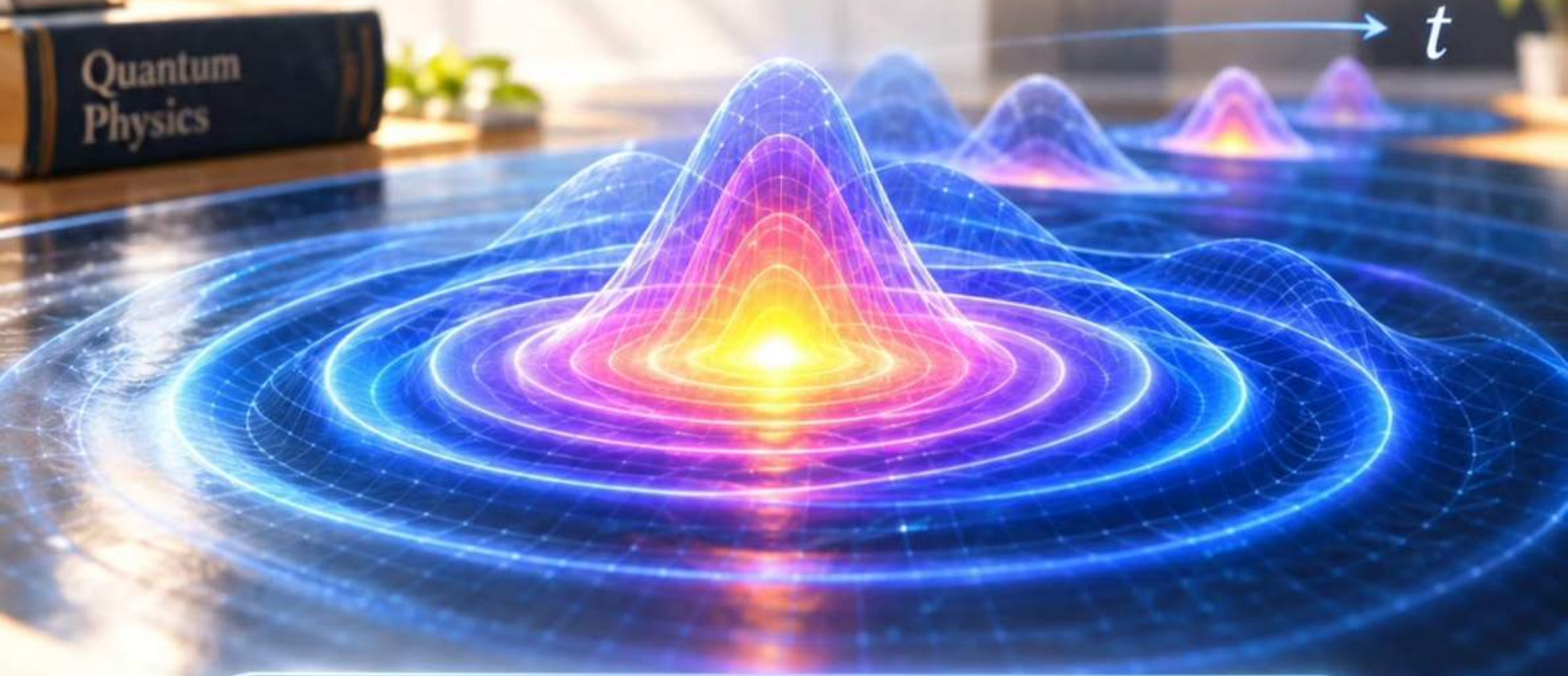
This term links spatial variation and mass to **kinetic energy.**

The Schrödinger Equation

A simple visual guide for everyone

$$i\hbar \frac{\partial \Psi}{\partial t} = \left[\frac{-\hbar^2}{2m} \nabla^2 + V \right] \Psi$$

It tells us how a **quantum** state changes over time.



Swipe to understand every part step by step →

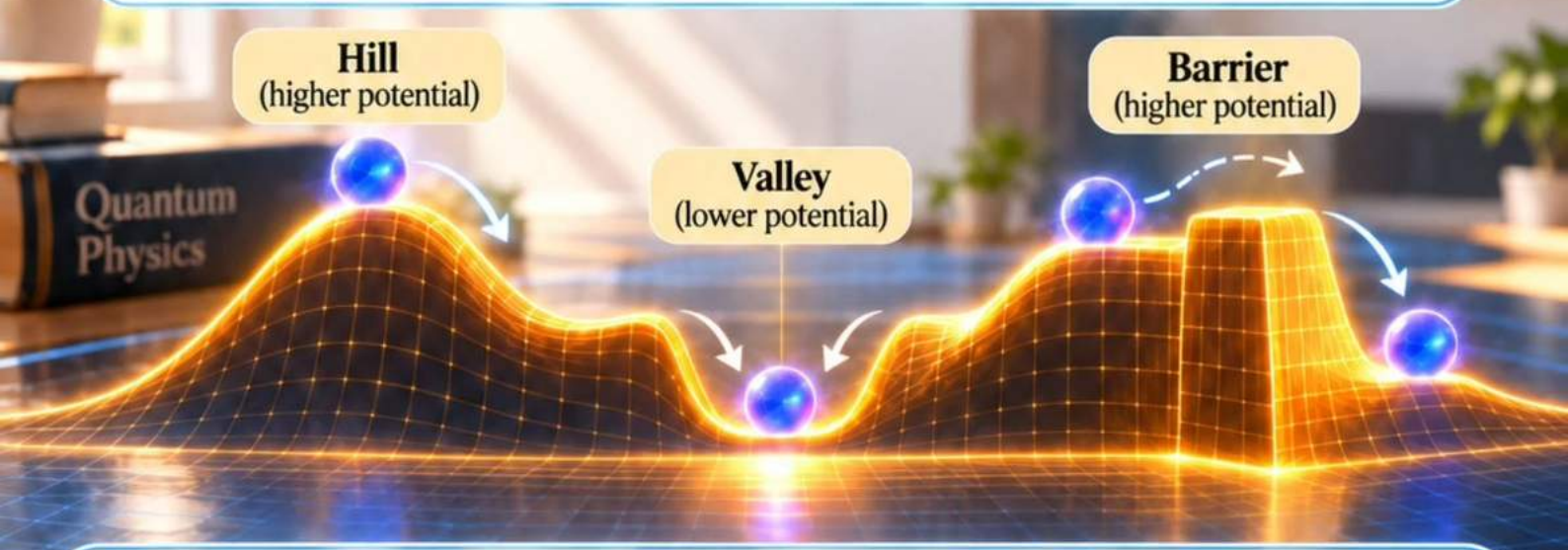
V: the effect of the surroundings

V stands for **potential energy.**

It describes the **environment** around the particle.

That can include **electric fields, barriers, wells, or traps.**

It shows how the surroundings **influence the wavefunction.**



Analogy: V **is like the shape of a landscape.**

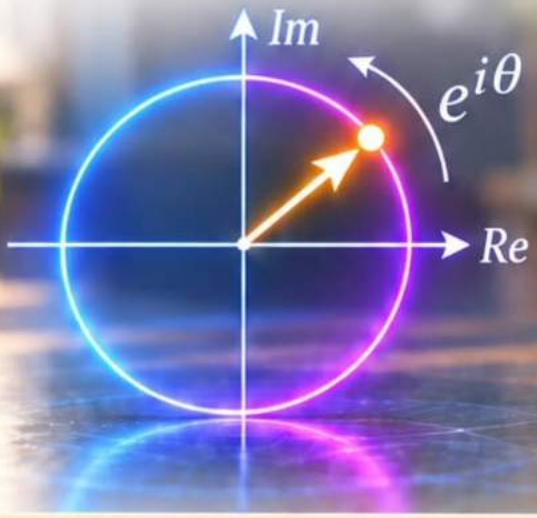
Hills, valleys, and walls show where motion is easier or harder.
It is an analogy for energy, not a literal path.

Potential energy is an energy landscape the particle feels.

i and $\hbar$: two small symbols with big jobs

i

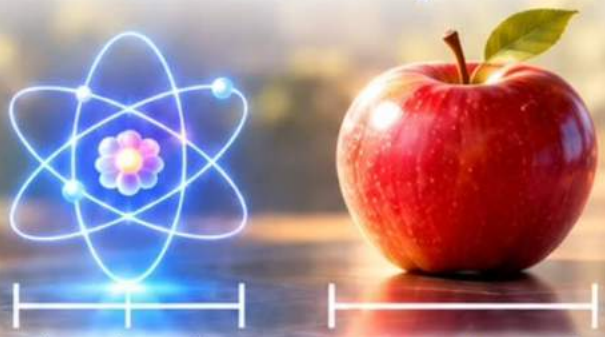
- i is the imaginary unit.
- It keeps track of phase in the wavefunction.
- It lets the math describe oscillations and interference.



Analogy: think of i as the marker for the wave's phase or timing.

$\hbar$ *h-bar*

- $\hbar$ is the reduced Planck constant.
- It sets the natural scale of quantum effects.
- It tells us we are dealing with the microscopic world.



Atomic scale
$\sim 10^{-10}$ m
(0.1 nanometer)

Everyday scale
$\sim 10^{-1}$ m
(10 centimeters)

Analogy: think of $\hbar$ as a built-in ruler for the microscopic world.

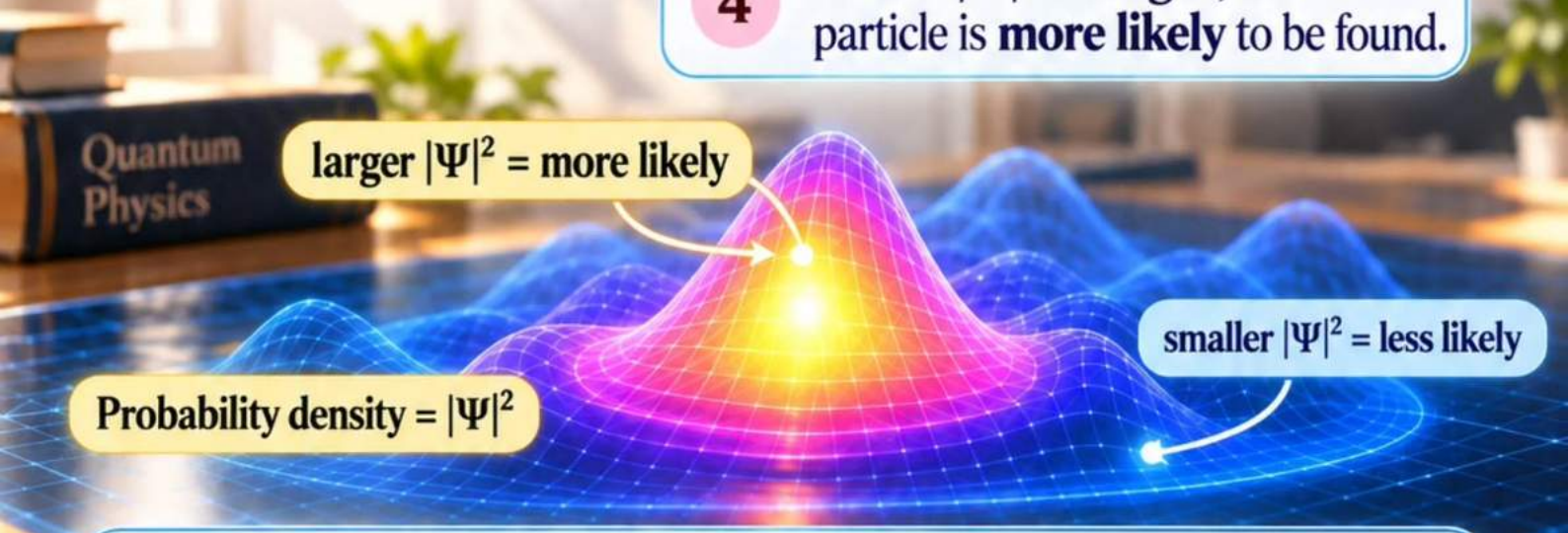
i tracks phase, and $\hbar$ sets the quantum scale.

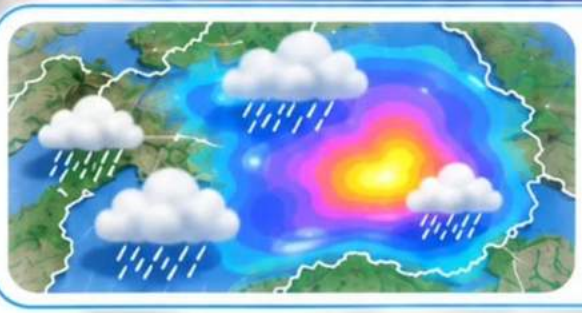
Ψ: the wavefunction

A map of possibilities

Ψ

1. Ψ describes the **quantum state** of the particle.
2. It **does not give one exact position.**
3. Instead, it encodes **all the possible outcomes.**
4. Where $|\Psi|^2$ is **larger**, the particle is **more likely** to be found.





Analogy: like a weather map showing where rain is **more likely,** not a promise of one exact raindrop.

Ψ helps determine the probability pattern.

$\partial\Psi/\partial t$: change over time

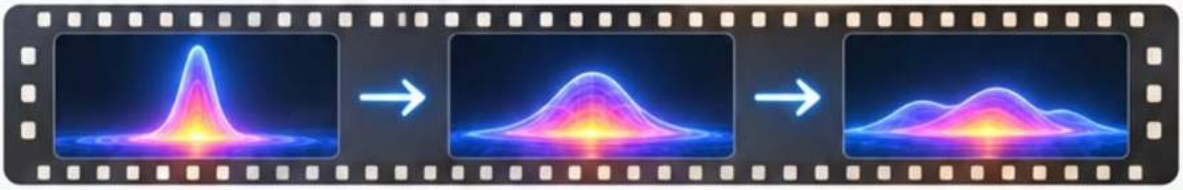
How fast the wavefunction is changing

1. This term asks: **how *is* Ψ changing right now?**
2. Is the probability wave **growing, shrinking, moving,** or **spreading?**
3. It focuses on the **rate of change** at this moment.





Analogy: like comparing frames of a video to see how a moving shape changes from one moment to the next.

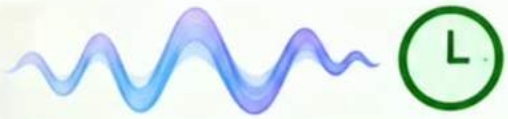


This is the **'time-change'** part of the equation.

What is this equation *really* saying?

$$i\hbar \frac{\partial \Psi}{\partial t} = \left[\frac{-\hbar^2}{2m} \nabla^2 + V \right] \Psi$$

Left side:
how the quantum state is **changing right now.**

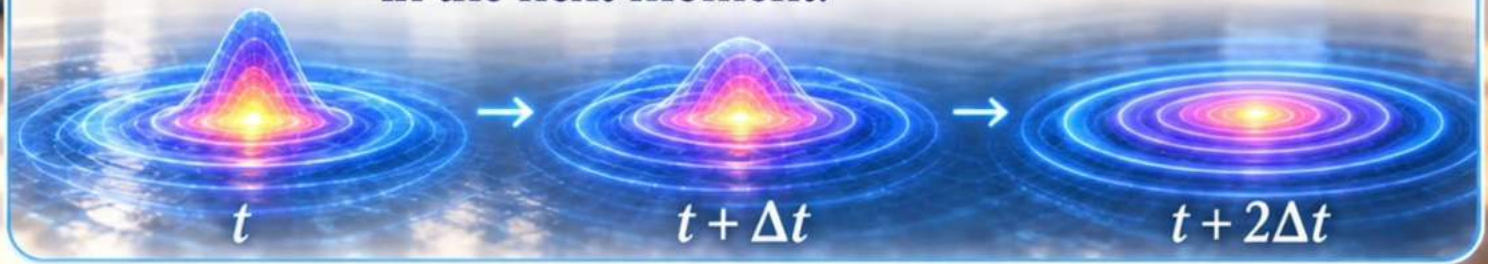


Right side:
what is **making it change.**



In short: the equation predicts how a particle's probability wave **evolves over time.**

Analogy: like predicting how ripples on water will change in the next moment.



This is the **main equation of motion** in quantum mechanics.

Putting it all together

$$i\hbar \frac{\partial \Psi}{\partial t} = \left[\frac{-\hbar^2}{2m} \nabla^2 + V \right] \Psi$$

Ψ describes the **quantum state.**

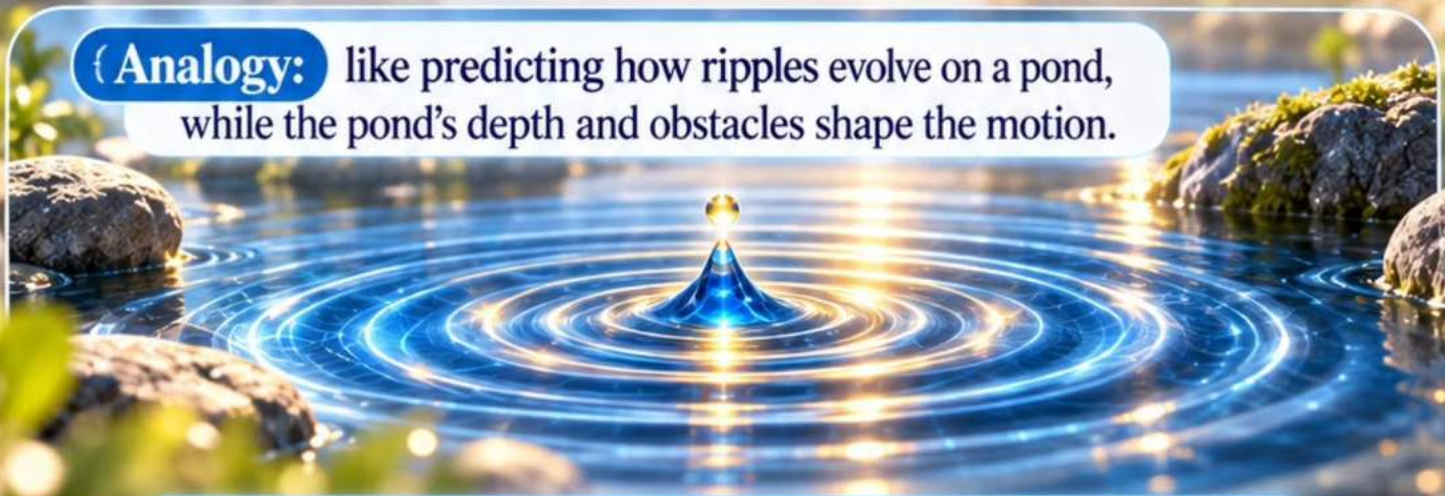
$\frac{\partial \Psi}{\partial t}$ tells how it is **changing in time.**

$\frac{-\hbar^2}{2m} \nabla^2$ The **kinetic term** depends on the wave's shape and the particle's mass.

V V adds the effect of the **surroundings.**

The **Schrödinger equation** predicts how a particle's probability wave changes over time because of its shape, its mass, and its **environment.**

Analogy: like predicting how ripples evolve on a pond, while the pond's depth and obstacles shape the motion.



This is why the equation is central to quantum mechanics.

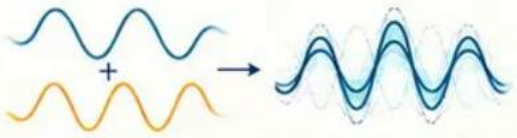
HOW TO STUDY QUANTUM MECHANICS

A concept-first roadmap • From zero to advanced

01 FOUNDATION

Energy • Momentum • Waves • Interference

Learn how waves reinforce, cancel, and form standing patterns.



02 PHOTON ENERGY

$E = hf$

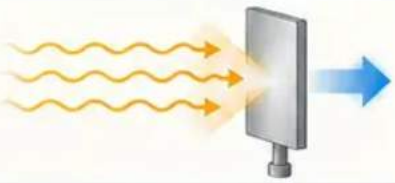
Frequency sets energy per photon.
At fixed frequency, brighter light means more photons.



03 PHOTON MOMENTUM

$p = h/\lambda$

Light transfers momentum.
Shorter wavelength means greater photon momentum.



04 MATTER WAVES

$\lambda = h/p$

Electrons diffract.
A matter wave is not a wavy particle trajectory.



05 WAVEFUNCTION & PROBABILITY

$\psi \rightarrow |\psi|^2$

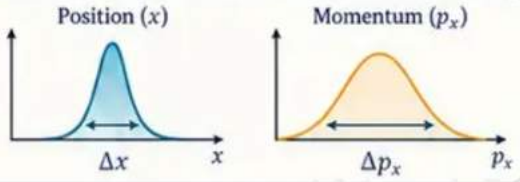
Amplitudes interfere; individual detections are localized.



06 UNCERTAINTY

$\Delta x \, \Delta p_x \geq \hbar/2$

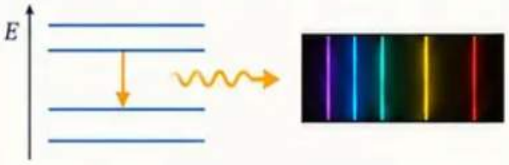
A narrower position distribution requires a wider momentum distribution.



07 BOHR: A HISTORICAL BRIDGE

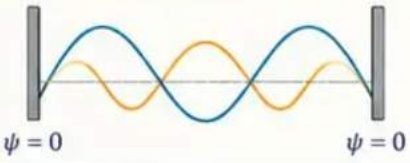
$|\Delta E| = hf$

Discrete energies explain spectral lines.
Definite electron orbits are not the modern picture.



08 CONFINEMENT & QUANTIZATION

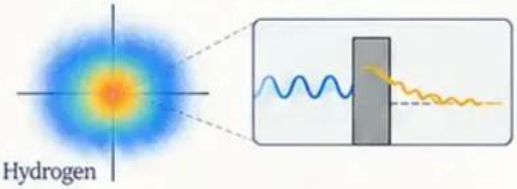
Boundary conditions select allowed wave patterns and bound-state energies.
Unconfined systems can have continuous energies.



09 SCHRÖDINGER EQUATION

$i\hbar \, \partial\psi/\partial t = \hat{H}\psi$

Predict how states evolve and find stationary states.
Study: Free particle → Wells → Tunnelling → Oscillator → Hydrogen



10 ADVANCED PATH



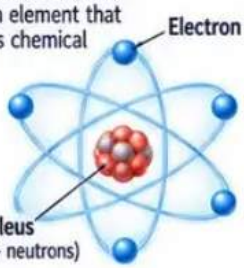
Math alongside physics: Calculus • Complex numbers • Linear algebra • Probability

ATOMIC STRUCTURE

15 KEY POINTS

1 What is an Atom?

An atom is the smallest unit of an element that retains its chemical identity.



2 Subatomic Particles

Particle	Symbol	Charge	Mass (approx.)	Location
Proton	p^+	+1	1 u	Nucleus
Neutron	n	0	1 u	Nucleus
Electron	e^-	−1	1/1836 u	Electron cloud

3 Atomic Number and Mass Number

Z = number of protons

$$A = Z + N$$

$$N = A - Z$$

For a neutral atom: number of electrons = Z

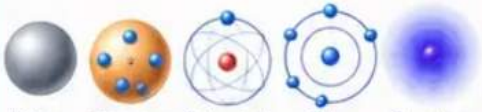
4 Isotopes, Isobars, Isotones

Isotopes: same Z, different N
^{12}C and ^{14}C

Isobars: same A, different Z
^{14}C and ^{14}N

Isotones: same N, different Z
^{14}C and ^{15}N

5 Historical Atomic Models



Dalton (1803) · Thomson (1904) · Rutherford (1911) · Bohr (1913) · Quantum model (1920s)

6 Bohr's Theory (Selected Formulas)

Energy level:

$$E_n = -\frac{13.6}{n^2}\ \text{eV}$$

Radius of orbit:

$$r_n = a_0 n^2$$

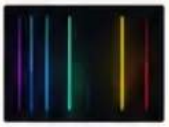
(a_0 = 0.529 Å)

7 Atomic Spectra (Rydberg Equation)

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$R_H = 1.097 \times 10^7\ m^{-1}$

$n_2 > n_1$



Hydrogen emission spectrum

8 Wave–Particle Duality (de Broglie)

$$\lambda = \frac{h}{p}$$



For a nonrelativistic particle:

$$\lambda = \frac{h}{mv}$$

9 Heisenberg Uncertainty Principle

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}$$

It is impossible to know the exact position and momentum of an electron simultaneously.

10 Quantum Numbers

Name	Symbol	Meaning	Values
Principal	n	Shell (energy level)	1, 2, 3, …
Azimuthal	l	Subshell (shape)	0 to $n-1$
Magnetic	m_l	Orbital orientation	$-l$ to $+l$
Spin	m_s	Electron spin	+½ or −½

11 Atomic Orbitals



s ($l = 0$) 1 orbital · p ($l = 1$) 3 orbitals · d ($l = 2$) 5 orbitals · f ($l = 3$) 7 orbitals

Subshell	No. of orbitals	Max. electrons
s	1	2
p	3	6
d	5	10
f	7	14

12 Rules of Electronic Configuration

1. **Aufbau principle** — Electrons fill lower-energy orbitals first.
2. **Pauli exclusion principle** — Maximum of 2 electrons per orbital with opposite spins.
3. **Hund's rule** — Electrons occupy degenerate orbitals singly with parallel spins before pairing.

13 Electronic Configuration (Examples)

H	(1)	$1s^1$
He	(2)	$1s^2$
C	(6)	$1s^2\ 2s^2\ 2p^2$
O	(8)	$1s^2\ 2s^2\ 2p^4$
Na	(11)	[Ne] $3s^1$
Cl	(17)	[Ne] $3s^2\ 3p^5$
Fe	(26)	[Ar] $3d^6\ 4s^2$
Cu	(29)	[Ar] $3d^{10}\ 4s^1$

14 Effective Nuclear Charge

$$Z_{eff} \approx Z - S$$

- Outer electrons experience reduced attraction due to shielding by inner electrons.
- Explains trends in atomic radius, ionization energy and electron affinity.

15 Important Formulas (Atomic Structure)

Concept	Formula
Mass number	$A = Z + N$
Shell capacity	$2n^2$
No. of orbitals in shell	n^2
Photon energy	$E = h\nu$
Photon wavelength	$\lambda = c/\nu$
de Broglie wavelength	$\lambda = h/p$
Hydrogen energy levels	$E_n = -\left(\frac{13.6}{n^2}\right)$ eV
Bohr radius (H atom)	$r_n = a_0 n^2$
Uncertainty principle	$\Delta x\, \Delta p_x \geq \hbar/2$
Effective nuclear charge	$Z_{eff} \approx Z - S$